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Effects of Deletion Neighbourhood Frequency and Individual Differences in Lexical Decision, Progressive Demasking, and Naming

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This study investigates whether deletion neighbourhood frequency influences the processing of French written words, and whether it might also be influenced by individual differences in skilled adult readers. For this purpose, words with at least 1 higher-frequency deletion neighbour (e.g., *pliage* [*folding*]/*plage* [*beach*]) and others with no higher-frequency neighbour (e.g., *morose* [*gloomy*]) were presented in lexical decision (Experiment 1), progressive demasking (Experiment 2), and naming (Experiment 3) tasks. For each experiment, the participants' lexical skills were assessed by spelling, reading, and vocabulary tests. In Experiments 1–3, participants responded more slowly to words with at least 1 higher-frequency deletion neighbour than they did to words with no such neighbour. We also found evidence that the inhibitory effect of deletion neighbourhood frequency was sensitive to lexical skills in the naming task. These findings are discussed in terms of lexical competition underlying visual word recognition according to individual differences.

Public Significance Statement

This study indicated that visual word recognition depends on both orthographic similarity between words and individual differences in adults' lexical skills (spelling, reading, and vocabulary). We found that French readers responded more slowly to words with at least 1 higher-frequency deletion neighbour (e.g., *pliage* [*folding*]/*plage* [*beach*]) than they did to words with no such neighbour. This effect occurred for all French readers in lexical decision and identification tasks and emerged for high lexical-skill readers in naming. These findings suggest that lexical competition between words is a critical process for the French language.

Keywords: visual word recognition, individual differences, lexical inhibition, deletion neighbours, orthographic neighbourhood

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The cognitive processes underlying visual word recognition are still a critical issue in psycholinguistics. Competition between orthographically similar word representations (i.e., orthographic neighbours) is a core assumption in current models of visual word recognition (e.g., Chen & Mirman, 2012; Davis, 2010; Grainger & Jacobs, 1996), and it is important to determine whether individual differences among skilled readers emerge in sensitivity to lexical inhibition in different languages (see Andrews, 2015) and in different experimental tasks. Recently, the role of individual differences among skilled readers has become a key question for researchers in psycholinguistics and in visual word recognition, which represents a challenge for the models that traditionally consider implicitly that visual word recognition relies on the same underlying mechanisms for all adult readers (see Andrews, 2012, 2015) However, literature investigating a combination of orthographic neighbourhood effects and lexical skills is still scarce and restricted to the English language. The present study uses three word-processing tasks to investigate whether lexical skills influence one of the most-studied orthographic neighbourhood effects in the French language, the neighbourhood frequency effect, which is assumed to rely on lexical competition.

The Orthographic Neighbourhood Frequency Effect in Visual Word Recognition

Research in psycholinguistics has extensively examined the lexical competition processes underlying visual word recognition by comparing words with at least one higher-frequency orthographic neighbour than the printed word, and words with no such neighbour (see Andrews, 1997; Grainger, 2008; Mathey, 2001; Perea, 2015, for reviews). Many studies have investigated the effects of a word's orthographic neighbours that differ by substi-

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tuting one letter (e.g., *car/cat*; Coltheart, Davelaar, Jonasson, & Besner, 1977), hereafter referred to as substitution neighbours. In their seminal study, Grainger, O'Regan, Jacobs, and Segui (1989) showed that French words with at least one higher-frequency substitution neighbour were identified more slowly and less accurately than words with no higher-frequency substitution neighbour in the lexical-decision task. This inhibitory neighbourhood frequency effect has been replicated in different languages (e.g., in Spanish, Carreiras, Perea, & Grainger, 1997; in French, Mathey & Zagar, 2006; in Dutch, van Heuven, Dijkstra, & Grainger, 1998, in English; Perea & Pollatsek, 1998), although the inhibitory neighbourhood frequency effect appeared with greater difficulty in English (e.g., Sears, Campbell, & Lupker, 2006). This effect was also found in other tasks frequently used to investigate word processing, such as the progressive demasking task (e.g., in Spanish, Carreiras, et al., 1997; in French, Grainger & Jacobs, 1996; in English, McArthur, Sears, Scialfa, & Sulsky, 2015), and the naming task (in Spanish, Carreiras, et al., 1997).

However, considering only substitution neighbours has proved to be too restrictive to investigate the effects of orthographic similarity between words because it only takes into account words of the same length differing by one letter with no change in letter positions, and several researchers have proposed extended operationalization (e.g., Andrews, 1996; Davis, Perea, & Acha, 2009; Yarkoni, Balota, & Yap, 2008). Specifically, the inhibitory neighbourhood frequency effect has been demonstrated in English words differing by the transposition of two adjacent letters (e.g., *salt/slat*; Andrews, 1996), referred to as transposition neighbours, in both lexical decision and naming tasks. This effect has also been shown in the lexical-decision task with words differing by the addition or deletion of one letter; respectively, referred to as addition neighbours (e.g., *jugar* [*play*]/*juzgar* [*judge*]) and deletion neighbours (e.g., *consejo* [*advice*]/*conejo* [*rabbit*]), both in English and in Spanish (Davis et al., 2009; Davis & Taft, 2005).

Since the study by Grainger et al. (1989), the neighbourhood frequency effect can be considered as an index of lexical competition during visual word recognition in the framework of interactive activation and competition models (McClelland & Rumelhart, 1981; see also Chen & Mirman, 2012; Davis, 2010). Indeed, a core assumption of such models is competition between coactivated representations of similar words. In this theoretical framework, when a printed word is presented, lexical representations of orthographically similar words are preactivated. These coactivated candidates compete with each other by means of a lateral inhibition mechanism at the word level, until the word stimulus representation is activated enough to reach the identification threshold. When an orthographic neighbour is of higher frequency than the stimulus word, it is strongly activated and competition takes longer; thus, slowing down the recognition of the stimulus word. Although the interactive activation model proposed by McClelland and Rumelhart (1981) helps account for competition effects from activated representations of higher-frequency substitution neighbours, it does not predict any activation of transposition neighbours, addition neighbours, and deletion neighbours, because of a strict coding of letter position for words of the same length (e.g., Davis et al., 2009; Grainger, 2008). The spatial coding model of visual word identification (Davis, 2010) can be used to explain the effects of such extended orthographic neighbourhoods. In this model, activation of the orthographic neighbour involves a method called superposition matching, to make the activated lexical representation match the presented stimulus. Moreover, letter representation is coded regardless of its position within the word, but according to an activation gradient, corresponding to a positional gradient. In this way, the spatial coding model captures the neighbourhood frequency effect for all types of neighbourhoods (Davis, 2010).

The Influence of Individual Differences on Orthographic Neighbourhood Effects

Visual word recognition is not only sensitive to lexical factors, but is also influenced by individual differences among skilled readers (e.g., Perfetti, 2007). A few studies have also examined whether the effects of higher-frequency orthographic neighbours also depend on individual differences (Andrews, 2012, 2015 for reviews). In the lexical-decision task, Sears et al. (2006) examined whether substitution neighbours of higher frequency than the stimulus words could influence the recognition of these words, compared with words with no higher-frequency substitution neighbour in English. They also investigated the possible influence of individual reading ability differences on the effect. Individual differences were assessed using the Author Recognition Test (Stanovich & West, 1989) in which participants have to decide whether or not names correspond to real authors. The results showed an effect because of individual differences, with slower response times (RTs) and fewer errors for the group with a high reading ability than for the group with a low reading ability. However, no evidence was found for either the inhibitory neighbourhood frequency effect in English or for the possibility that individual differences in reader ability could influence this effect. As previous research had mostly demonstrated the inhibitory neighbourhood frequency effect in languages other than English, such as French (e.g., Grainger et al., 1989), Spanish (e.g., Carreiras et al., 1997), and Dutch (van Heuven et al., 1998), Sears et al. (2006) proposed an explanation in terms of language differences and task sensitivity to inhibition. They argued that it was difficult for the inhibitory neighbourhood frequency effect to appear in the standard lexical-decision task in English because of the English neighbourhood structure, a structure that requires weaker inhibitory connections within the lexicon. They also assumed that the effect would be easier to detect with other experimental paradigms, such as masked orthographic priming, which is more sensitive to lexical inhibition in English because the neighbours are strongly preactivated when they are presented as primes (e.g., Davis & Lupker, 2006). Finally, it could be argued that the measure of individual differences in reading ability used by Sears et al. (2006) may not have been sufficient to be predictive of changes in the neighbourhood frequency effect. Indeed, according to Perfetti (2007), the measure of reading skills does not necessarily include a correct retrieval of words with their orthographic representation; other skills, such as spelling, or vocabulary, should also be measured.

More recently, Andrews and Lo (2012) administered reading, spelling, and vocabulary tests to identify differences in lexical skills and examined their influence in a primed lexical-decision task. The results showed inhibitory priming effects with transposition neighbour and substitution neighbour word primes. Differences were also found in word processing according to individual differences in lexical skills. A stronger inhibitory priming effect was found for both transposition neighbour and substitution neighbour primes only in individuals who had a higher general score of lexical skills based on reading, spelling, and vocabulary measures. The data suggest that lexical competition effects occurred for skilled readers with high-quality lexical representations (see also Andrews & Hersch, 2010 for similar results on neighbourhood density), which was in line with the lexical quality hypothesis (Perfetti, 2007). Nevertheless, the role of individual differences in word processing among skilled adult readers remains to be examined in languages other than English.

The Present Study

The general purpose of this study is to investigate lexical competition processes underlying visual word recognition along with their sensitivity to adult readers' lexical skills in the French language. To achieve this, we examined the orthographic deletion neighbourhood frequency effect in French through three experiments, using the lexical decision (Experiment 1), the progressive demasking (Experiment 2), and the naming tasks (Experiment 3). These tasks, which are the most widely used in research in visual word recognition, are considered to involve both common processes (i.e., lexical access), and task-specific processes (see, e.g., Ferrand et al., 2011). Investigating the neighbourhood frequency effect across these three tasks may help to shed more light on the issue of whether lexical competition depends on individual differences in lexical skills. Therefore, in Experiments 1–3, we compared words with at least one higher-frequency deletion neighbour (e.g., *pliage* [*folding*]/*plage* [*beach*]) with words with no higher-frequency deletion neighbour (e.g., *morose* [*gloomy*]). We used deletion neighbours because such orthographic neighbours differ in length compared with the stimulus word. This neighbourhood type also provides an understanding of the sensitivity of word length in visual recognition (Davis et al., 2009), and may be particularly sensitive to individual differences in readers' lexical precision. In each experiment, we used different tests to measure the French participants' levels of spelling, reading, and vocabulary to assess their lexical skills and subsequently estimate their lexical quality (see Andrews & Lo, 2012, in English). Differences in readers' lexical skills would imply variation in lexical precision (speed and accuracy of word identification), and lexical quality may contribute to efficiency in lexical competition processes underlying word identification (see Andrews, 2015; Andrews & Lo, 2012 for priming experiments). Therefore, individual differences among skilled readers could emerge in the lexical decision, progressive demasking, and naming tasks.

Experiment 1: Lexical Decision

The aim of Experiment 1 was to test the effect of deletion orthographic neighbourhood frequency for French words presented in the lexical-decision task, which is the most widely used task in the field of visual word recognition, and also to determine whether individual differences in readers' lexical skills could influence this effect. Although the inhibitory deletion neighbourhood frequency effect has never been shown in French, we assumed that it would occur, as it has been observed in both English and Spanish with the lexical-decision task (Davis et al., 2009).

To our knowledge, no study has investigated the combined issue of orthographic neighbourhood and lexical skills of adult readers for the French language. We assumed that individuals with low lexical skills would have difficulties with word identification: The lower individuals' lexical skills, the higher RTs and error rates. Furthermore, as it was argued for the English language to explain individual differences in the inhibitory effect of orthographic priming with substitution neighbour and transposition letter primes (Andrews & Hersch, 2010; Andrews & Lo, 2012), we assumed that the individual variations in lexical precision could lead to differences in the establishment of lexical competition from the deletion neighbourhood in the lexical-decision task in French. More precisely, the inhibitory effect of higher-frequency deletion neighbourhood should increase with the score of lexical skills.

Method

Participants. The participants were 92 volunteer undergraduate students in human sciences (91% in psychology) from the University of Bordeaux. They were aged from 18 to 30 years (M = 20.65, SD = 2.18). All were French native speakers and reported having normal or corrected-to-normal vision.

Measures of individual differences. The participants were assessed on six measures of spelling, reading, and vocabulary skills assumed to provide an estimation of their lexical skills, using a method inspired by the one proposed by Andrews and Lo (2012) and adapted to the French language.

Spelling skills. Two spelling tests from "Evaluation des Compétences de Lecture chez l'Adulte de plus de 16 ans" (ECLA-16; Gola-Asmussen, Lequette, Pouget, Rouyet, & Zorman, 2011) were administered. The first test consisted of a dictation of words (regular and irregular ones) and pseudowords. We noted RTs in seconds for each word. The second test consisted of a dictation of a text of 83 words and four sentences, requiring grammar and syntax knowledge. We calculated a score out of 20 (see Gola-Asmussen et al., 2011).

Reading skills. Participants had to read two texts aloud. The first one was "Alouette R," a text with frequent and rare words grouped in grammatically and syntactically correct sentences, but with little meaning (LeFavrais, 2005). Participants were asked to read it as quickly as they could and try not to make mistakes. A reading speed score was collected. The second text was "Le Pollueur" (*The Polluter*) from ECLA-16 (Gola-Asmussen et al., 2011), a text with meaning. Participants were asked to read it as quickly as they could, and try not to make mistakes, for 1 min. The score was the number of correctly read words.

Vocabulary. Participants were assessed on vocabulary skills using two tests. The first one was the Mill Hill Vocabulary (Synonyms; Raven, French adaptation from Deltour, 1993), which consisted of 34 items. For each item, participants had to choose the synonym from six choices. The number of correct responses was collected. The second test was the LexTALE vocabulary test in French (Brysbaert, 2013), which consisted in presenting 56 French words and 28 French-looking nonwords. Participants were asked to choose the real words from the 84 stimuli. The Ghent score was calculated for this test (see Brysbaert, 2013).

Finally, the dictation RTs (M = 45, SD = 4.9), the dictation scores (M = 17.4, SD = 1.6), the Alouette R reading speed scores (M = 549, SD = 81.1), the "*Le Pollueur*" reading accuracy scores (M = 192, SD = 19.7), the Mill Hill vocabulary scores (M = 32.1, SD = 4.7), and the Ghent vocabulary scores (M = 43.9, SD = 4.9) were transformed into standardized scores. We then averaged a composite score of lexical skills for each participant (see Figure 1 for a frequency distribution of this score in our sample).

Stimuli. The 120 stimulus words (see online supplementary materials, Table 1 of Appendix A) were selected for the lexical-decision task from the French lexical database Lexique (New, Brysbaert, Veronis, & Pallier, 2007; New, Pallier, Brysbaert, & Ferrand, 2004). All words were 5–7 letter words, and none of them had any substitution neighbour. They were divided into two experimental conditions according to deletion neighbourhood frequency: 60 words had at least one higher-frequency deletion neighbour (e.g., *pliage* [*folding*]/*plage* [*beach*]), and 60 others had no higher-frequency deletion neighbour (e.g., *morose* [*gloomy*]). In the latter condition, the words had at least one lower-frequency deletion neighbour to control the number of deletion neighbours. The stimuli were also matched across the two conditions based on several other lexical factors, including lexical frequency, number of letters, number of syllables number of substitution neighbours, and Orthographic Levenshtein Distance (Yarkoni et al., 2008; see Ferrand et al., 2010 for the French language). The main statistical characteristics of the word materials are presented in Table 1. To provide "no" trials for the lexical-decision task, we created 120 pronounceable pseudowords. They were matched on length with the stimulus words and all of them had at least one deletion neighbour to ensure that the task required lexical categorisation.

Procedure. The participants were tested individually in a quiet room after providing written informed consent. They were administered two spelling tests, two reading tests, and two vocabulary tests. They then performed the lexical-decision task run by the DMDX display software (Forster & Forster, 2003). Each trial began by displaying a fixation cross "+" for 500 ms, followed by the stimulus in grey, lower case 12-point Courier New at the centre of a black computer screen, until the participant responded. All participants saw the 240 stimuli in a different random order. Participants were instructed to decide whether the stimulus was a French word or not, as quickly and as accurately as possible by pressing one button with their dominant hand if the stimulus was a word, and another button with their other hand if it was a pseudoword. The experiment began after a sample of 16 practice trials. RTs and accuracy were recorded after each trial.

Table 1
Mains Characteristics of the Word Materials and of Their Higher-Frequency Deletion Neighbours

Variables	Deletion neighbourhood frequency: HFDN	Deletion neighbourhood frequency: No HFDN	p Value
Stimulus word			
Example	*pliage*	*morose*	
Lexical frequency	3.97 (*9.7*)	4.65 (*4.6*)	ns
Bigram frequency	9494	8970	ns
Number of letters	6.40	6.40	ns
Number of syllables	2.18	2.18	ns
Number of DNs	1.43	1.28	ns
Number of HFDNs	1.30	.00	$p < .001$
Number of SNs	.00	.00	—
OLD 20	1.97	2.03	ns
HFDN			
Example	*plage*	—	
Lexical frequency	33.02	—	$p < .001$

Note. HFDN = higher-frequency deletion neighbor; Lexical frequency = mean word film and word book frequencies in number of occurrences per million; OLD 20 = Orthographic Levenshtein Distance; ns = not significant ($p > .10$). Standard deviations are in parentheses.

Results

RTs below 300 ms or above 1,500 ms (see also Carreiras et al., 1997) were excluded from the analyses (1.64% of the data). Eight items were removed because of a high error rate (more than 41%). The data were entered into separate generalised linear mixed models (GLMMs) of correct RTs and error rates to test the effects of crossed and nested subject and item factors. GLMMs are a combination of linear mixed models and generalised linear models,

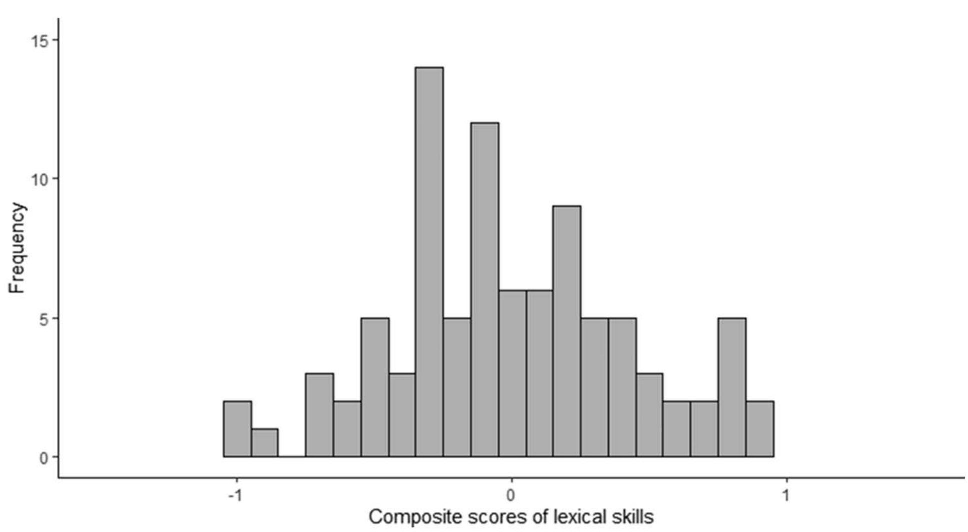

Figure 1. Frequency distribution of the composite scores of lexical skills in Experiment 1 (lexical-decision task).

relaxing the normality distribution assumption of RTs and residuals, and avoiding transformations of RTs (i.e., log; see Lo & Andrews, 2015). We introduced the Composite Scores of Lexical Skills (continuous), and Deletion Neighbourhood Frequency (at least one higher-frequency deletion neighbour vs. no higher-frequency deletion neighbour) as fixed factors, and participants and items as random factors. All models were fitted using the χ^2 log likelihood ratio tests with maximum likelihood parameter estimation (see Appendix B, Table B1 in the online supplementary materials for more details about the models). Figure 2 shows how RTs vary as a function of composite scores of lexical skills and the deletion neighbourhood frequency.

The GLMM analysis with gamma distribution and a log link function of correct word RTs showed a significant main effect of deletion neighbourhood frequency, $\chi^2(1) = 8.93$, $p = .003$. RTs were 33 ms longer for words with at least one higher-frequency deletion neighbour than for words with no higher-frequency deletion neighbour. There was also a main effect of the composite scores of lexical skills, $\chi^2(1) = 8.18$, $p = .004$. Figure 1 shows that the higher the composite scores of lexical skills, the lower the RTs ($b = -.042$, $SE = .015$). The interaction between the composite scores of lexical skills and the deletion neighbourhood frequency was not significant, $\chi^2(1) = 1$, $p = .31$.

We conducted a GLMM analysis of the error rates with a binomial variance assumption applied to the binary data. The analysis also showed that the trend main effect of deletion neighbourhood frequency was not significant, $\chi^2(1) = 2.73$, $p = .098$. The main effect of the composite scores of lexical skills was significant, $\chi^2(1) = 27.89$, $p < .001$. The higher the composite scores of lexical skills, the lower the error rates ($b = -.889$, $SE = .168$). The interaction between the composite scores of lexical skills and deletion neighbourhood frequency was not significant, $\chi^2(1) < 1$.

Discussion

Experiment 1 investigated the processes underlying the effect of higher-frequency deletion neighbours in the lexical-decision task, according to individual differences in lexical quality among French skilled readers. The main findings can be summarised as follows. First, words with at least one higher-frequency deletion neighbour were recognised more slowly than words with no such neighbour. Second, lexical skills were related to RTs and accuracy: The higher lexical skills, the lower RTs, and error rates. Third, the inhibitory deletion neighbourhood frequency effect on word decision times did not differ reliably according to lexical skills. These results suggest that lexical competition occurs for all individuals during visual word recognition of French words in the lexical-decision task.

The first finding is that we replicated the inhibitory effect of higher-frequency deletion neighbours that was previously obtained in English and Spanish in the lexical-decision task (Davis et al., 2009), and extended it to the French language. The results showed that lexical decision latencies were longer were higher for words with at least one higher-frequency deletion neighbour (e.g., *pliage* [*folding*]/*plage* [*beach*]) than for words with no higher-frequency deletion neighbour (e.g., *morose* [*gloomy*]). These findings can be interpreted within the spatial coding model of visual word identification (Davis, 2010). In this interactive activation theoretical framework, a scheme of spatial coding is used for coding letter position dynamically, so that the letters are encoded independently of the position they have in the word with a positional gradient. The letter position uncertainty may or may not reinforce letter activation at the correct position within the word. Thus, through this process, higher-frequency deletion neighbours of the stimulus word are activated when the printed word is presented and become efficient competitors during visual word recognition. When the inhibition mechanism is achieved by a method of superposition matching between the stimulus and the competitor candidate, the printed word can be recognised. In this case, the correct "yes" response would be delayed compared with that for words with no such deletion neighbour. The lexical matching would be affected by the degree of letter position uncertainty coding. Therefore, this experiment provides initial evidence of a deletion neighbourhood frequency effect in French, indicating that visual word recognition

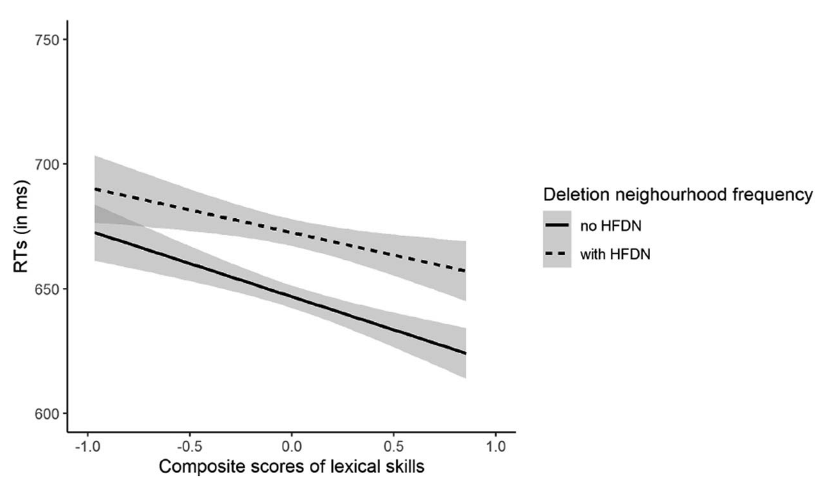

Figure 2. Correct word RTs as a function of deletion neighbourhood frequency and composite scores of lexical skills in Experiment 1 (lexical-decision task). Shaded error bands represent 95% confidence intervals. HFDN = higher-frequency deletion neighbour.

in the French language is sensitive not only to substitution neighbours, but also to deletion neighbours. Future research in French (and other languages) should, therefore, be attentive to this type of orthographic neighbourhood.

The second finding of our study is that individual differences in the lexical skills of French adults are related to RTs and accuracy in the lexical-decision task. Following the approach developed by Andrews (see Andrews, 2015) for the English language, a combined measure of spelling, reading, and vocabulary scores was used to assess French readers' lexical precision. Our results indicate that the lower individuals' lexical skills, the longer word recognition latencies, and the higher error rates, thereby replicating previous data in English (Andrews & Hersch, 2010; Andrews & Lo, 2012; Sears et al., 2006). This finding is also consistent with the lexical quality hypothesis (Perfetti, 2007), stating that strong lexical representations lead to fast and accurate retrieval of these representations, contrary to weak lexical representations. Thus, individuals with low lexical quality may develop weak lexical precision, encountering inefficiency in the visual word recognition processes (see Andrews & Lo, 2012). Along with the results of previous studies, these data promote the incorporation of lexical skills in research in psycholinguistics in different languages. Not all readers have the same lexical skills, and consequently word identification performances can differ and might even rely on different lexical access processes (see Andrews, 2015).

The third finding of this study is that the deletion neighbour frequency effect did not vary according to individuals' lexical skills. Contrary to, Andrews' research in English with the masked priming paradigm (Andrews & Hersch, 2010; Andrews & Lo, 2012), our data with the standard lexical-decision task do not suggest that lexical competition differs according to individual differences in lexical skills in French. As we expected, we only observed inhibition (and no facilitation as reported in English) from the higher-frequency neighbour in French. Words with a higher-frequency deletion neighbour were classified more slowly in the lexical-decision task than words with no higher-frequency deletion neighbour. However, the increase of inhibition with lexical skills we expected did not emerge. Previous results from a primed lexical-decision task in English have indicated that the orthographic priming effect was inhibitory for readers with high lexical skills while it was facilitatory for readers with low lexical skills (Andrews & Hersch, 2010), suggesting that lexical competition only occurs in readers with high lexical skills (see also Andrews & Lo, 2012). Language differences could explain this difference. According to Sears et al. (2006), the lateral inhibition process may be less dominant in English than in other languages such as French, probably because there are more neighbours and higher-frequency neighbours in the English language; thus, requiring weaker lateral inhibitory connections in the lexicon to achieve lexical access. Contrary to the findings from the English language indicating that lexical competition does not arise for all readers (e.g., Andrews & Hersch, 2010; Andrews & Lo, 2012), our data suggest that the inhibition mechanism during word competition arises for all skilled French readers, at least in the lexical-decision task. To go further, we examined the deletion neighbourhood frequency effect in a progressive demasking task, in which the visual degradation of the stimuli would make it possible to specify this effect and also potentially reveal individual differences in word identification processes.

Experiment 2: Progressive Demasking

This study examined the differences in lexical skills in inhibition processes in visual word recognition using the progressive demasking task. To our knowledge, no published experiment has used this task to study lexical competition arising from deletion neighbours. Yet, the progressive demasking task would be particularly sensitive to lexical competition (Carreiras et al., 1997; Grainger & Jacobs, 1996; McArthur et al., 2015). This task would make the identification of the stimulus uncertain, which would lead participants to use early information to give an answer, and consequently make it possible to select an incorrect candidate (McArthur et al., 2015). This task can be used to conduct a qualitative analysis of errors, insofar as participants type the word. Thus, the higher-frequency neighbour may be selected and reported, rather than the word presented (Grainger & Segui, 1990, with substitution neighbours). Furthermore, the progressive demasking task requires the explicit identification of the stimulus (Dufau, Stevens, & Grainger, 2008; Ferrand et al., 2011). Thus, a correct answer requires the explicit resolution of the competition between the higher-frequency orthographic neighbour and the word stimulus (Andrews, 1996). As previously shown for substitution neighbours in the progressive demasking task (Carreiras et al., 1997) and in keeping with our lexical-decision task data for deletion neighbours (Experiment 1), we predicted that a higher-frequency deletion neighbourhood inhibitory effect should occur for RTs and error rates in the progressive demasking task. To go further, our words with no higher-frequency deletion neighbour had at least one lower-frequency deletion neighbour (to control for deletion neighbour density), so we assumed that the generated responses would correspond to the higher-frequency deletion neighbour more than to the lower-frequency deletion neighbour. As for the lexical-decision task used in Experiment 1, we also predicted an effect of individual differences in lexical skills in the progressive demasking task. Both faster RTs and lower error rates were expected with the increase of individuals' lexical skills. Finally, we investigated whether the deletion neighbourhood frequency effect is related to individuals' lexical skills in the progressive demasking task.

Method

Participants. The participants were 92 new volunteer undergraduate students in human sciences (92% in psychology) from the University of Bordeaux. They were aged from 18 to 30 years ($M = 20.85$, $SD = 2.41$). All were French native speakers and reported having normal or corrected-to-normal vision.

Measures of individual differences. The participants were evaluated on their lexical skills using the same six tests as in Experiment 1. The dictation RTs ($M = 46$, $SD = 7.7$), the dictation scores ($M = 16.3$, $SD = 2.2$), the Alouette R reading speed scores ($M = 523$, $SD = 91.2$), the Le Pollueur reading accuracy scores ($M = 185$, $SD = 24.5$), the Mill Hill vocabulary scores ($M = 31.5$, $SD = 4.5$), and the Ghent vocabulary scores ($M = 42.1$, $SD = 5.4$) were transformed into standardized scores. We then averaged a composite score of lexical skills for each participant (see Figure 3 for a frequency distribution of this score in Experiment 2).

Stimuli. The stimulus words were the same as in Experiment 1.

Procedure. The participants were assessed individually on their spelling, reading and vocabulary levels following the same

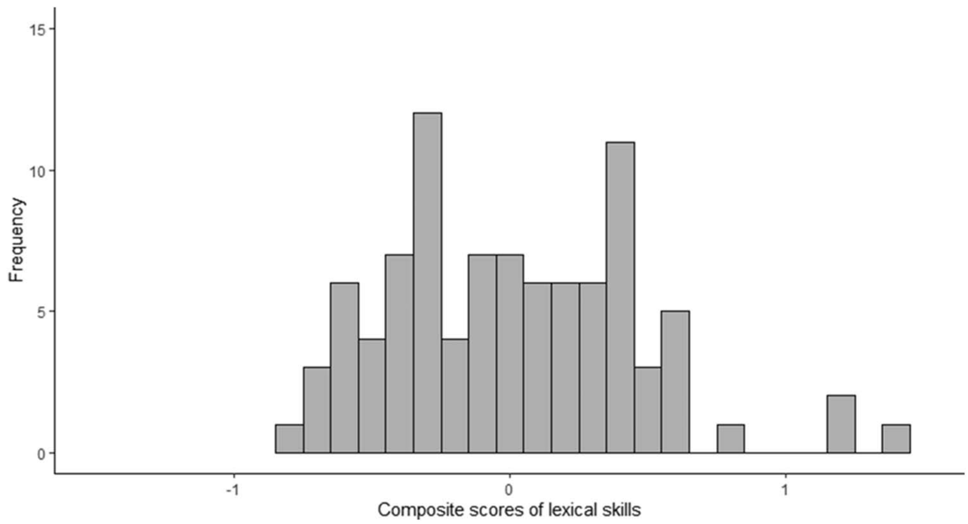

Figure 3. Frequency distribution of the composite scores of lexical skills in Experiment 2 (progressive demasking).

procedure as in Experiment 1. They then performed the progressive demasking task run with software by Dufau et al. (2008). Each trial consisted of an alternation between the stimulus word in lowercase and the mask (i.e., a row of hash marks #### of the same length as the stimulus) in the centre of the computer screen. The display time for the word gradually increased from 15 ms in 15 ms increments, while the display duration of the mask gradually decreased from 195 ms in 15 ms increments. Thus, on the first presentation, the word was presented for 15 ms and the mask for 195 ms. On the second presentation, the word was presented for 30 ms and the mask for 180 ms. The procedure was permitted to continue to its full extent until the participant's response, without an intervening mask following 14 alternations. RTs were measured from the beginning of the first cycle until the participant's response. The participants had to stop the demasking sequence as soon as they thought that they had identified the word by pressing the "space" key on the keyboard, and then typing the word. They were instructed to respond as quickly and as accurately as possible. The task started with 10 practice words, followed by the 120 experimental words in a different random order for each participant.

Results

Analysis of correct RTs and error rates. RTs below 300 ms or above 3,500 ms (see also Carreiras et al., 1997) were excluded from the analyses (1.89% of the data). One item was removed because of a high error rate (more than 41%). Correct RTs and error rates were submitted to separate GLMMs, with the Composite Scores of Lexical Skills (continuous), and Deletion Neighbourhood Frequency (at least one higher-frequency deletion neighbour vs. no higher-frequency deletion neighbour) as fixed factors, and participants and items as random factors. All models were fitted using the χ^2 log likelihood ratio tests with maximum likelihood parameter estimation (see Appendix B, Table B2 in the online supplementary materials for more details about the models). Figure 4 shows how RTs vary as a function of composite scores of lexical skills and the deletion neighbourhood frequency.

The GLMM analysis with gamma distribution and a log link function of correct word RTs showed a significant main effect of deletion neighbourhood frequency, $\chi^2(1) = 1.02$, $p < .001$. RTs were 222 ms longer for words with at least one higher-frequency deletion neighbour than for words with no higher-frequency deletion neighbour. There was also a main effect of composite scores of lexical skills, $\chi^2(1) = 8.02$, $p < .001$. Figure 2 shows that the higher the composite scores of lexical skills, the lower the RTs ($b = -.129$, $SE = .014$). Finally, the interaction between the composite scores of lexical skills and deletion neighbourhood frequency was not significant, $\chi^2(1) = 1.64$, $p = .20$.

We conducted a GLMM analysis on the error rates with a binomial variance assumption applied to the binary data. The analysis also showed that the main effect of deletion neighbourhood frequency was significant, $\chi^2(1) = 7.39$, $p = .006$. The percentage of errors was higher for words with at least one higher-frequency deletion neighbour ($M = 11.4\%$, $SE = .43$) than for words with no higher-frequency deletion neighbour ($M = 7.3\%$, $SE = .34$). The main effect of the composite scores of lexical skills was not significant, $\chi^2(1) = 1.84$, $p = .17$, and this latter variable did not interact with deletion neighbourhood frequency, $\chi^2(1) = 1.34$, $p = .24$.

Analysis of error rates corresponding to deletion neighbours. The progressive demasking task allowed us to perform a finer error analysis based on qualitative responses. In the experimental condition of words with no higher-frequency deletion neighbour, all stimulus words had at least one lower-frequency deletion neighbour (to control deletion neighbourhood density across the two experimental conditions). Thus, in the set of errors for the task, we identified the errors corresponding to deletion neighbours, that is, when the participants typed a deletion neighbour instead of the printed word (e.g., *plage* [*beach*] written instead of *pliage* [*folding*]).

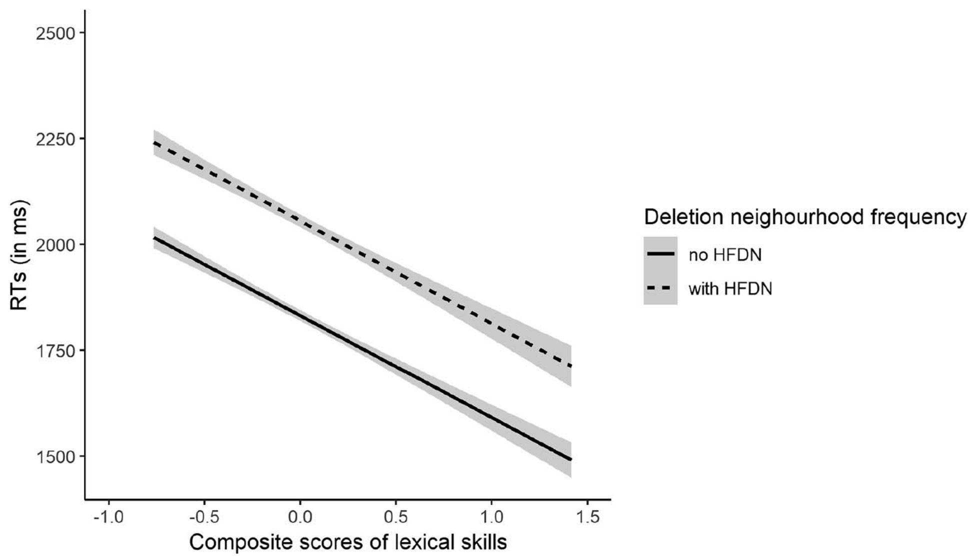

Figure 4. Correct word RTs as a function of deletion neighbourhood frequency and composite scores of lexical skills in Experiment 2 (progressive demasking). Shaded error bands represent 95% confidence intervals. HFDN = higher-frequency deletion neighbour.

We conducted a GLMM analysis on the error rates corresponding to deletion neighbours. The analysis also showed that the main effect of deletion neighbourhood frequency was significant, $\chi^2(1) = 23.29$, $p < .001$. The percentage of errors was higher for words with at least one higher-frequency deletion neighbour (M = 3.4%, SE = .24) than for words with no higher-frequency deletion neighbour (M = 0.4%, SE = .08). The main effect of the composite scores of lexical skills was significant, $\chi^2(1) = 4.64$, p = .031. The higher the composite scores of lexical skills, the lower the error rates corresponding to deletion neighbour ($b = -.905$, $SE = .419$). Finally, the interaction between composite scores of lexical skills and deletion neighbourhood frequency was not significant $\chi^2(1) = 2.32$, $p = .12$.

Discussion

The aim of Experiment 2 was to examine whether the deletion neighbourhood frequency effect also occurs in the progressive demasking task, and how individual differences in lexical skills could influence this effect. In addition, as the progressive demasking task can be used to perform a qualitative analysis of the errors, we expected that the effects predicted for correct word RTs and error rates would also emerge in error rates restricted to the deletion neighbours. The data showed that the expected inhibitory deletion neighbourhood frequency effect was found on these three measures (i.e., RTs, error rates, and error rates corresponding to deletion neighbours increased for words with at least one higher-frequency deletion neighbour), as did the expected effect of the lexical skills in two measures (i.e., RTs and error rates corresponding to deletion neighbours decreased as lexical skills increased). Finally, no interaction between the frequency of deletion neighbours and the composite scores of lexical skills was found in the progressive demasking task.

The first finding of this experiment is that the inhibitory deletion neighbour frequency effect found in the lexical-decision task (Experiment 1) was replicated in the progressive demasking task. Moreover, we extended the effect previously observed with substitution neighbour (e.g., Carreiras et al., 1997) to deletion neighbour for the first time. In the progressive demasking task, the presentation of the stimulus word in a visual degraded condition could prompt the reader to rely on early process information, like letter features and letter processes (Grainger, 2008). In the spatial coding model (Davis, 2010), letter activation processes may be more difficult to operate because of this visual degradation, making letter position and letter identity coding uncertain for potential candidates. Following activation from the letter level, the representations of the stimulus and of its competitors would be coactivated and lateral inhibition would set in via the lexical matching mechanism. The competitors would then either be inhibited, slowing down word recognition, or cause erroneous responses (McArthur et al., 2015). Our results showed that the erroneous responses restricted to deletion neighbours occurred more often for higher-frequency deletion neighbours (e.g., typing *plage* [*beach*] instead of *pliage* [*folding*]), than for lower-frequency deletion neighbours that were hardly ever reported. This confirms that the higher-frequency deletion neighbour was sufficiently activated to be identified instead of the stimulus word. In this case, lateral inhibition would not have resolved the competition between the two words representations.

The second new finding of Experiment 2 is that the individuals' lexical skills were related to the progressive demasking task response speed: The higher individuals' lexical skills, the lower RTs. These results are consistent with those of the standard lexical-decision task (Experiment 1) and those of the primed lexical-decision task (see Andrews, 2015 for a review). According to the lexical quality hypothesis (Perfetti, 2007), lexical representations are less stable, less precise in the memory of individuals with low lexical skills, so they would make it more difficult to retrieve a word's representation.

Finally, we did not find the expected interaction between the deletion neighbourhood frequency and lexical skill group either for correct word RTs or error rates, as in Experiment 1. One possible explanation is that the use of the progressive demasking task may have favoured the development of lexical competition processes, probably because the final identification time is considerably increased in this task because of the successive mask presentations (Grainger & Segui, 1990). Therefore, in this task lexical competition occurs for all French readers. In both lexical decision and progressive demasking tasks, spelling aspects of words should also be used to perform the task (e.g., Ferrand et al., 2011). To go further, we could investigate the efficiency of the lexical representation retrieval in the naming task, which requires more phonological processes.

Experiment 3: Naming

The aim of this experiment was to study how higher-frequency deletion neighbours could influence the reader's naming performance in the naming task, according to differences in lexical skills. Previous naming studies have mostly shown a facilitatory effect of substitution neighbour density (see Andrews, 1997; Mathey, 2001 for reviews). This facilitation in reading aloud would be because of the phonological activation of the word and its neighbour representations. The neighbourhood frequency effect has been less investigated than the neighbourhood density effect in the naming task. With substitution neighbours, the neighbourhood frequency effect was found to be either facilitatory (Carreiras et al., 1997; Grainger, 1990) or inhibitory (Carreiras et al., 1997). According to Carreiras et al. (1997), the direction of the effect would depend on neighbourhood density: Facilitation was reported with a high substitution neighbour density (i.e., M = 9.6 in Grainger, 1990, in Dutch; M = 7.5 in Carreiras et al., 1997, in Spanish) while inhibition was found when substitution neighbour density was low (i.e., M = 2.1; in Carreiras et al., 1997, in Spanish). Consistently, the transposition neighbour frequency effect reported by Andrews (1996) was inhibitory, when the density of transposition neighbours was low (i.e., M = 4.25 in English). The stimuli of our study had a low density of deletion neighbours ranging from 1 to 3 (M = 1.33, see Table 1). Therefore, in keeping with the results of Carreiras et al. (1997) and Andrews (1996) for substitution neighbours and transposition neighbours, respectively, we expected an inhibitory higher-frequency deletion neighbour effect in the naming task. In addition, we predicted that naming times would increase with individuals' lexical skills. Indeed, according to the lexical quality hypothesis (Perfetti, 2007), the strength of the links between the different identities of words (phonological, orthographic, and semantic) is responsible for the quality of lexical representations. Individuals with low lexical skills would have weak links between the spelling and the phonology of the word, which are components mobilized in the naming task. Accordingly, lexical competition would be more developed for participants with high- than for low-lexical skills, so that we expected the deletion neighbour frequency effect to increase with the participants' lexical skills.

Method

Participants. The participants were 90 new volunteer students in Human Sciences (93% in psychology) from the University of Bordeaux. They were aged from 18 to 33 years (M = 21.3, SD = 3.6). They were French native speakers and reported having normal or corrected-to-normal vision.

Measures of individual differences. The participants were evaluated on their lexical skills using the same six tests as in Experiment 1. The dictation RTs (M = 48, SD = 7.3), the dictation scores (M = 16.8, SD = 2.2), the Alouette R reading speed scores (M = 517, SD = 82.4), the Le Pollueur reading accuracy scores (M = 181, SD = 22.9), the Mill Hill vocabulary scores (M = 32, SD = 4.5), and the Ghent vocabulary scores (M = 43.6, SD = 4.3) were transformed into standardized scores. We then averaged a composite score of lexical skills for each participant (Figure 5 for a frequency distribution of this score in Experiment 3).

Stimuli. The stimulus words were the same as in Experiment 1.

Procedure. After signing informed consent, the participants were assessed on their lexical skills using the same spelling,

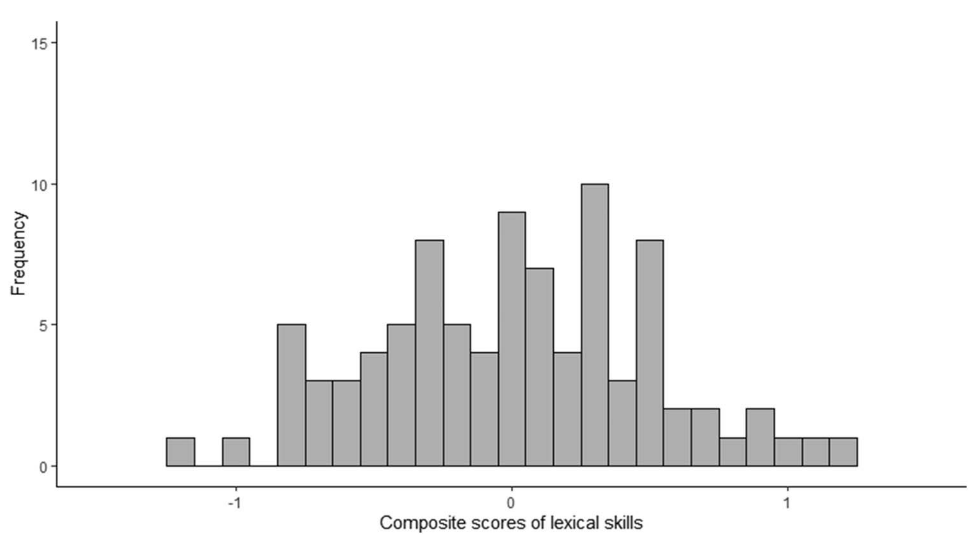

Figure 5. Frequency distribution of the composite scores of lexical skills in Experiment 3 (naming).

reading, and vocabulary tests as in Experiment 1. The naming task was run with the E-prime software (Schneider, Eschman, & Zuccolotto, 2002). The words appeared (New Courier, font 18, white on a black background) in the centre of a computer screen. The order in which the words were presented was randomized for each participant, so that none of the participants saw the words in the same order. Each trial consisted of displaying a fixation cross for 1,500 ms, followed by a word until the participant began to speak in a microphone connected to a SRbox, which was connected to the computer. The participants had to read the words aloud in the microphone, as quickly and accurately as possible. Naming times were recorded using the software. Errors from the participant (e.g., mispronunciations, incorrect words, and hesitations) were noted by the experimenter. The experiment began with 10 practice words, followed by the 120 experimental words.

Results

RTs below 300 or above 800 ms (see also Carreiras et al., 1997) were excluded from the analyses (5.69% of the data). Naming times and error rates were submitted to separate GLMMs, with the Composite Scores of Lexical Skills (continuous), and Deletion Neighbourhood Frequency (at least one higher-frequency deletion neighbour vs. no higher-frequency deletion neighbour) as fixed factors, and participants, items, and the phonological categories of the first phoneme of each word (see Balota, Cortese, Sergent-Marshall, Spieler, & Yap, 2004) as random factors. All models were fitted using the χ^2 log likelihood ratio tests with maximum likelihood parameter estimation (see Appendix B, Table B3 in the online supplementary materials for more details about the models). Figure 6 shows how RTs vary as a function of composite scores of lexical skills and the deletion neighbourhood frequency.

The GLMM analysis with Gaussian distribution and a log link function of correct word RTs showed a significant main effect of deletion neighbourhood frequency, $\chi^2(1) = 29.34$, $p < .001$. RTs were 6 ms longer for words with at least one higher-frequency deletion neighbour than for words with no higher-frequency deletion neighbour. The main effect of the composite scores of lexical skills was not significant, $\chi^2(1) < 1$. The interaction between the composite scores of lexical skills and the deletion neighbourhood frequency was significant, $\chi^2(1) = 7.02$, $p = .008$. Figure 3 shows that the higher the composites scores of lexical skills, the greater the inhibitory deletion neighbourhood frequency effect.

Finally, we conducted a GLMM analysis on the error rates with a binomial variance assumption applied to the binary data. The analysis also showed that the main effect of deletion neighbourhood frequency was significant, $\chi^2(1) = 6.03$, $p = .014$. The percentage of errors was higher for words with at least one higher-frequency deletion neighbour ($M = 5.7\%$, $SE = .32$) than for words with no higher-frequency deletion neighbour ($M = 3.1\%$, $SE = .23$). The main effect of the composite scores of lexical skills was significant, $\chi^2(1) = 11.32$, $p < .001$. The higher the composite scores of lexical skills, the lower the error rates ($b = -.567$, $SE = .168$). Finally, the interaction between the composite scores of lexical skills and deletion neighbourhood frequency was significant, $\chi^2(1) = 5.09$, $p = .02$. The lower the composite scores of lexical skills, the higher the inhibitory deletion neighbourhood frequency effect ($b = -.456$, $SE = .202$).

Discussion

The aim of Experiment 3 was to examine the influence of deletion neighbourhood frequency in the naming task, as well as whether it is sensitive to individual differences in lexical skills. The main findings can be summarised as follows. First, we found the expected inhibitory higher-frequency deletion neighbour effect on both correct word RTs and error rates. Second, the results showed that lexical skills were related to error rates. Finally, an interaction between deletion neighbour and lexical skills occurred in both speed and accuracy data.

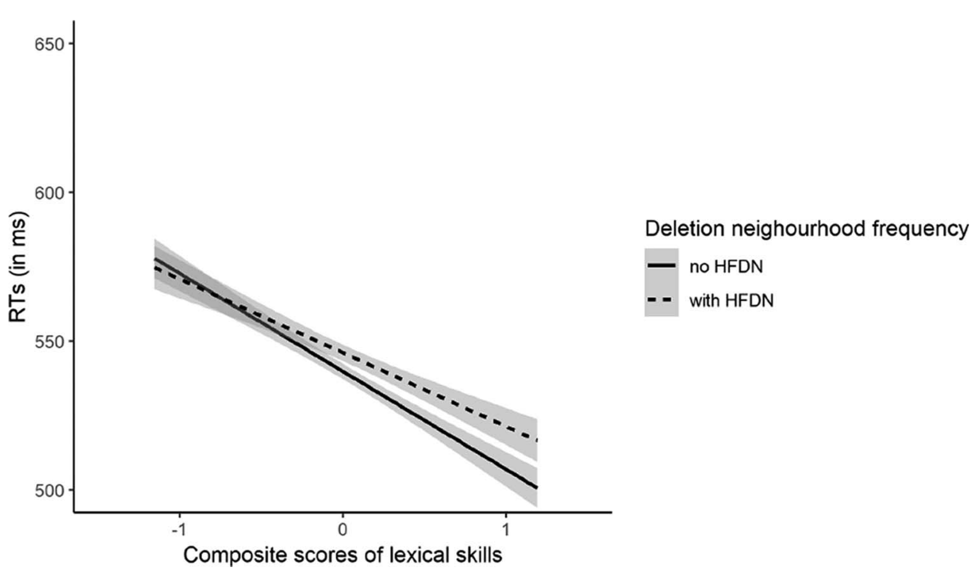

Figure 6. Correct word RTs as a function of deletion neighbourhood frequency and composite scores of lexical skills in Experiment 3 (naming). Shaded error bands represent 95% confidence intervals. HFDN = higher-frequency deletion neighbour.

As shown in both the lexical decision (Experiment 1) and the progressive demasking tasks (Experiment 2), the naming task data revealed an inhibitory effect of higher-frequency deletion neighbourhood. The naming times were longer and the accuracy was lower for words with at least one higher deletion neighbour than for words with no such neighbour. These data show for the first time that the inhibitory neighbourhood effect of deletion neighbours previously found in the lexical-decision task for English and Spanish words (Davis et al., 2009; Davis & Taft, 2005) can be extended to the naming task and to the French language. As expected, the effect of deletion neighbours was inhibitory, which is consistent with previous findings found for transposition neighbours and substitution neighbours in naming when neighbourhood density was low, in both English and Spanish (Andrews, 1996; Carreiras et al., 1997).

Facilitatory effects of higher-frequency neighbourhoods reported in previous naming studies (Andrews, 1989, 1992, 1997; Grainger, 1990) could rather be attributed to a high number of orthographically similar words that would also have similar pronunciations; therefore, partially activating the stimulus word representation. In that case, the phonological component would facilitate the word process in the naming task (e.g., see Carreiras et al., 1997). In our experiment, we found an inhibitory higher-frequency deletion neighbour effect, suggesting that the phonological component was not activated enough to facilitate the pronunciation of the stimulus word. The orthographic component, and in particular lexical competition between the stimulus word representation and those of its orthographic neighbours of higher frequency, would be responsible for the effect. The interactive activation model (McClelland & Rumelhart, 1981) can explain such lexical competition effects for substitution neighbours, but not for transposition neighbours (Andrews, 1996) or deletion neighbours that are not even supposed to be activated, because of the strict coding of letter position in this model. According to Davis (2010), although the spatial coding model was initially designed for visual word recognition and has no phonological output units, it can be used to explain naming data provided by a lexical route. Therefore, we assume that in this framework the inhibitory neighbourhood effect we obtained in the naming task relies on the same competition processes as the effects found in both the lexical decision and progressive demasking tasks.

Our data show that individual differences in lexical skills are related to the accuracy of the word naming task: The lower individuals' lexical skills, the higher error rates. We can assume that lexical processes would be less efficient, which seems to be consistent with the lexical quality hypothesis (Perfetti, 2007).

In addition, we found the predicted interaction between deletion neighbour frequency and lexical skills. The inhibitory higher-frequency deletion neighbour effect on word naming RTs emerged progressively when the level of lexical skills of the readers increased, so that only high lexical skill participants exhibited the effect. These results are consistent with those of the primed lexical-decision task reported in English (Andrews & Lo, 2012), suggesting that the processing of written words can differ according to lexical skills, in particular in terms of lexical competition. Words with at least one higher-frequency deletion neighbour were read more slowly than words with no such neighbour, and even more so for readers with high rather than low lexical skills. These results are also consistent with the view that the French language may require the implementation of a dominant inhibition process, contrary to English (Sears et al., 2006). Our data can be interpreted within the spatial coding model (Davis, 2010). Orthographic and phonological representations in individuals with high lexical skills would be sufficiently stable, strong, and precise to activate both candidates, unlike individuals with low lexical skills. The activated candidates in memory would then be compared with the presented stimulus, via the superposition matching method, making it possible to establish inhibition. Once the inhibition has been correctly put in place, the stimulus representation is recognised and pronounced correctly, which is more often the case for individuals with high lexical skills than for those with low skills. Lexical competition would develop to more extent with the increase of lexical skill readers. The error data suggested that individuals with low lexical skills have more difficulty decreasing and effectively resolving lexical competition from the higher-frequency deletion neighbour.

General Discussion

This study investigated whether the deletion neighbourhood frequency effect occurs for French words presented in three experimental tasks that are widely used in word processing research (lexical decision, progressive demasking, and naming tasks), and also sought to determine the possible influence of individual differences in lexical skills on this effect. Overall, we have demonstrated the inhibitory deletion neighbourhood frequency effect, and the effect of the individuals' lexical skills in the lexical decision (Experiment 1), the progressive demasking tasks (Experiment 2), and the naming task (Experiment 3). We found that the inhibitory deletion neighbourhood frequency effect varied in the naming task according to differences in lexical skills.

The Inhibitory Deletion Neighbourhood Frequency Effect

The results obtained in three different experimental word processing tasks (i.e., lexical decision, progressive demasking, and naming tasks) showed an inhibitory frequency deletion neighbour effect in the French language.[1] First, we have extended the effect initially found in the English and Spanish language in the lexical-decision task (e.g., Davis et al., 2009). Higher-frequency deletion neighbours appear to be efficient competitors in visual word recognition, slowing down word identification (Davis, 2010; Davis et al., 2009). Second, the effect found in the lexical-decision task was reproduced in both the progressive demasking and the naming tasks. Because these three tasks are considered to involve both common and task-specific processes (e.g., Carreiras et al., 1997; Ferrand et al., 2011), this finding suggests that lexical competition from deletion neighbourhood reflects lexical access processes common to the three tasks. The effect of deletion neighbours would, therefore, reflect automatic activation and inhibition processes at the word level arising from lexical candidates of different

[1] Supplementary analyses were performed with only monomorphemic items to check the influence of morphemic complexity in each experiment. The inhibitory deletion neighbourhood frequency effect was also found. The morphological complexity did not seem to influence the deletion neighbourhood frequency effect.

length than the stimulus, which can be accounted for by the spatial coding model (Davis, 2010; see also Davis et al., 2009).

The Effect of Lexical Skills

The results of Experiments 1–3 demonstrate that readers' lexical skills are related with word RTs in the lexical decision and progressive demasking tasks, and with accuracy in the lexical decision, progressive demasking, and naming tasks, showing that the higher individuals' lexical skills, the faster RTs and/or the less error rates. Taken together, these results suggest that individuals with low lexical skills have difficulty in identifying the word, certainly because of weak orthographic and phonological representations (Andrews, 2015; Perfetti, 2007), which are involved to a variable extent in the three tasks. Indeed, the lexical-decision task involves extracting information from orthographic representations stored in memory (without necessarily having to retrieve the precise information; Andrews, 1996). The progressive demasking task additionally requires the explicit retrieval of orthographic representations in memory (the participants have to type the words on the keyboard and not just classify them as existing words or not; Dufau et al., 2008). Finally, the naming task involves both the orthographic and phonological components of the words to read aloud (Andrews, 1996; Ferrand et al., 2011) to a great extent. The links between the different components would be more or less strong according the individuals' lexical skills. Thus, the different experimental tasks we used seem relevant to study the role of individual differences in lexical skills in written word processing.

Is the Deletion Neighbourhood Frequency Effect Sensitive to Lexical Skills?

The three experiments in this study provide converging evidence in favour of lexical competition from deletion neighbours of higher frequency in visual word recognition of French words. In the naming task, which strongly involves the activation of both orthographic and phonological word representations, we have also shown that the deletion neighbour frequency effect was related to individual differences in lexical skills. As expected, naming latencies revealed that the inhibitory deletion neighbour frequency effect emerged in relation with the increase of individuals' lexical skills. The higher-frequency deletion neighbour seems to be activated more rapidly by individuals with the higher lexical skills, which would modulate lexical competition (see Andrews & Lo, 2012 for a similar interpretation of orthographic priming variations in English). However, we failed to show that French readers' lexical skills influenced the deletion neighbour frequency effect in either the lexical decision or the progressive demasking tasks. This may be attributed to the fact that the responses are longer in these latter tasks compared with the naming task (see also Carreiras et al., 1997) giving time to develop lexical competition for all readers. In addition, in the progressive demasking task, it could be argued that the degraded visual presentation would force early activation of the neighbours; therefore, favouring inhibitory effects (see Grainger & Segui, 1990 for similar interpretation with substitution neighbours).

Conclusion

In summary, this article provides new evidence, not only that the deletion neighbourhood frequency effect occurs in the French language, but also that individual differences in lexical skills can modify this effect in the word naming task. Our main findings suggest that lexical competition in visual word recognition arises for all French skilled readers in lexical decision and progressive demasking tasks, but to a greater extent and more efficiently for individuals with high lexical skills than for individuals with low lexical skills in naming. Our data are also consistent with the view that French might be more sensitive than English to lexical competition (see Sears et al., 2006), given that lexical competition only occurred for good English spellers in experiments using a priming procedure, which is supposed to preactivate the neighbour and, therefore, favour competition (Andrews & Lo, 2012). Further research should be conducted on the effects of the different types of neighbourhood (e.g., substitution neighbour, deletion neighbour, and transposition neighbour) in several tasks and languages, to shed more light on the mechanisms underlying the development of lexical activation and inhibition processes in visual word recognition according to both language and individual differences.

Résumé

La présente étude vise à déterminer si la fréquence lexicale des mots voisins obtenus par suppression d'une lettre influe sur le traitement de mots français écrits, et si elle peut aussi être influencée par les différences individuelles entre les lecteurs adultes aguerris. À cette fin, des mots ayant au moins un mot voisin plus fréquemment utilisé obtenu par suppression d'une lettre (p. ex., *pliage/plage*), et d'autres mots sans mots voisins plus « usités » obtenus par suppression d'une lettre (p. ex., *morose*) ont été présentés dans le cadre : d'un processus de décision lexicale (expérience 1), de démasquage progressif (expérience 2), et de dénomination (expérience 3). Lors de chaque expérience, les compétences lexicales ont été évaluées dans le cadre de tests d'épellation, de lecture et de vocabulaire. Lors des expériences 1 à 3, les participants répondaient plus lentement aux mots ayant au moins 1 mot voisin plus usité obtenu par suppression d'une lettre que les mots n'ayant aucun mot voisin de ce type. Nous avons aussi établi que l'effet inhibiteur de la fréquence lexicale des mots voisins obtenus par suppression d'une lettre était sensible aux compétences lexicales lors de la tâche de dénomination. Ces constatations sont abordées en termes de concurrence lexicale fondamentale à la reconnaissance visuelle des mots en fonction des différences individuelles.

Mots-clés : reconnaissance visuelle des mots, différences individuelles, inhibition lexicale, mots-voisins obtenus par suppression d'une lettre, voisinage orthographique.

References

Andrews, S. (1989). Frequency and neighborhood effects on lexical access: Activation or search? *Journal of Experimental Psychology: Learning, Memory, and Cognition, 15,* 802–814. http://dx.doi.org/10.1037/0278-7393.15.5.802

Andrews, S. (1992). Frequency and neighborhood effects on lexical access: Lexical similarity or orthographic redundancy? *Journal of Experimental*

Psychology: Learning, Memory, and Cognition, 18, 234–254. http://dx.doi.org/10.1037/0278-7393.18.2.234

Andrews, S. (1996). Lexical retrieval and selection processes: Effects of transposed-letter confusability. *Journal of Memory and Language, 35,* 775–800. http://dx.doi.org/10.1006/jmla.1996.0040

Andrews, S. (1997). The effect of orthographic similarity on lexical retrieval: Resolving neighborhood conflicts. *Psychonomic Bulletin & Review, 4,* 439–461. http://dx.doi.org/10.3758/BF03214334

Andrews, S. (2012). Individual differences in skilled visual word recognition and reading: The role of lexical quality. In J. Adelman (Ed.), Visual word recognition Volume *2: Meaning and context, individuals and development* (pp. 151–172). Sussex: Psychology Press.

Andrews, S. (2015). Individual differences among skilled readers. In A. Pollatsek & R. Treiman (Eds.), *The Oxford handbook of reading* (pp. 129–148). New York, NY: Oxford University Press.

Andrews, S., & Hersch, J. (2010). Lexical precision in skilled readers: Individual differences in masked neighbor priming. *Journal of Experimental Psychology: General, 139,* 299–318. http://dx.doi.org/10.1037/a0018366

Andrews, S., & Lo, S. (2012). Not all skilled readers have cracked the code: Individual differences in masked form priming. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 38,* 152–163. http://dx.doi.org/10.1037/a0024953

Balota, D. A., Cortese, M. J., Sergent-Marshall, S. D., Spieler, D. H., & Yap, M. (2004). Visual word recognition of single-syllable words. *Journal of Experimental Psychology: General, 133,* 283–316. http://dx.doi.org/10.1037/0096-3445.133.2.283

Brysbaert, M. (2013). Lextale_FR a fast, free, and efficient test to measure language proficiency in French. *Psychologica Belgica, 53,* 23. http://dx.doi.org/10.5334/pb-53-1-23

Carreiras, M., Perea, M., & Grainger, J. (1997). Effects of orthographic neighborhood in visual word recognition: Cross-task comparisons. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 23,* 857–871. http://dx.doi.org/10.1037/0278-7393.23.4.857

Chen, Q., & Mirman, D. (2012). Competition and cooperation among similar representations: Toward a unified account of facilitative and inhibitory effects of lexical neighbors. *Psychological Review, 119,* 417–430. http://dx.doi.org/10.1037/a0027175

Coltheart, M., Davelaar, E., Jonasson, J. T., & Besner, D. (1977). Access to the internal lexicon. In S. Dornic (Ed.), *Attention and performance VI* (pp. 535–555). New York, NY: Academic Press.

Davis, C. J. (2010). The spatial coding model of visual word identification. *Psychological Review, 117,* 713–758. http://dx.doi.org/10.1037/a0019738

Davis, C. J., & Lupker, S. J. (2006). Masked inhibitory priming in English: Evidence for lexical inhibition. *Journal of Experimental Psychology: Human Perception and Performance, 32,* 668–687. http://dx.doi.org/10.1037/0096-1523.32.3.668

Davis, C. J., Perea, M., & Acha, J. (2009). Re(de)fining the orthographic neighborhood: The role of addition and deletion neighbors in lexical decision and reading. *Journal of Experimental Psychology: Human Perception and Performance, 35,* 1550–1570. http://dx.doi.org/10.1037/a0014253

Davis, C. J., & Taft, M. (2005). More words in the neighborhood: Interference in lexical decision due to deletion neighbors. *Psychonomic Bulletin & Review, 12,* 904–910. http://dx.doi.org/10.3758/BF03196784

Deltour, J. J. (1993). Echelle de vocabulaire Mill Hill de J. C. Raven: Adaptation française et normes comparées du Mill Hill et du Standard Progressive Matrices (PM38) [Mill Hill Vocabulary Scale by J. C. Raven: French adaptation and comparative standards of Mill Hill and Standard Progressive Matrices (PM38)]. *Manuel et Annexes*. Braine le Château. Belgique: Application des Techniques Modernes.

Dufau, S., Stevens, M., & Grainger, J. (2008). Windows executable software for the progressive demasking task. *Behavior Research Methods, 40,* 33–37. http://dx.doi.org/10.3758/BRM.40.1.33

Ferrand, L., Brysbaert, M., Keuleers, E., New, B., Bonin, P., Méot, A., . . . Pallier, C. (2011). Comparing word processing times in naming, lexical decision, and progressive demasking: Evidence from chronolex. *Frontiers in Psychology, 2,* 306. http://dx.doi.org/10.3389/fpsyg.2011.00306

Ferrand, L., New, B., Brysbaert, M., Keuleers, E., Bonin, P., Méot, A., . . . Pallier, C. (2010). The French Lexicon Project: Lexical decision data for 38,840 French words and 38,840 pseudowords. *Behavior Research Methods, 42,* 488–496. http://dx.doi.org/10.3758/BRM.42.2.488

Forster, K. I., & Forster, J. C. (2003). DMDX: A windows display program with millisecond accuracy. *Behavior Research Methods, Instruments, & Computers, 35,* 116–124. http://dx.doi.org/10.3758/BF03195503

Gola-Asmussen, C., Lequette, C., Pouget, G., Rouyet, C., & Zorman, M. (2011). *ECLA 16+: Évaluation des compétences de lecture chez l'adulte de plus de 16 ans* [Assessment tool for reading competence in adults over 16]. Grenoble: Université de Provence Aix-Marseille I-Cognisciences LSE Université Pierre Mendès.

Grainger, J. (1990). Word frequency and neighborhood frequency effects in lexical decision and naming. *Journal of Memory and Language, 29,* 228–244. http://dx.doi.org/10.1016/0749-596X(90)90074-A

Grainger, J. (2008). Cracking the orthographic code: An introduction. *Language and Cognitive Processes, 23,* 1–35. http://dx.doi.org/10.1080/01690960701578013

Grainger, J., & Jacobs, A. M. (1996). Orthographic processing in visual word recognition: A multiple read-out model. *Psychological Review, 103,* 518–565. http://dx.doi.org/10.1037/0033-295X.103.3.518

Grainger, J., O'Regan, J. K., Jacobs, A. M., & Segui, J. (1989). On the role of competing word units in visual word recognition: The neighborhood frequency effect. *Perception & Psychophysics, 45,* 189–195. http://dx.doi.org/10.3758/BF03210696

Grainger, J., & Segui, J. (1990). Neighborhood frequency effects in visual word recognition: A comparison of lexical decision and masked identification latencies. *Perception & Psychophysics, 47,* 191–198. http://dx.doi.org/10.3758/BF03205983

LeFavrais, P. (2005). *Alouette-R, Test d'analyse de la lecture et de la dyslexie* [Reading and dyslexia analysis test]. Paris: Editions du Centre de Psychologie Appliquée.

Lo, S., & Andrews, S. (2015). To transform or not to transform: Using generalized linear mixed models to analyse reaction time data. *Frontiers in Psychology, 6,* 1171. http://dx.doi.org/10.3389/fpsyg.2015.01171

Mathey, S. (2001). L'influence du voisinage orthographique lors de la reconnaissance des mots écrits [Influence of the orthographic neighbourhood in written words recognition]. *Canadian Journal of Experimental Psychology, 55,* 1–23. http://dx.doi.org/10.1037/h0087349

Mathey, S., & Zagar, D. (2006). The orthographic neighbourhood frequency effect in French: A letter-case manipulation study. *Canadian Journal of Experimental Psychology, 60,* 159–165. http://dx.doi.org/10.1037/cjep2006015

McArthur, A. D., Sears, C. R., Scialfa, C. T., & Sulsky, L. M. (2015). Aging and the inhibition of competing hypotheses during visual word identification: Evidence from the progressive demasking task. *Neuropsychology, Development, and Cognition Section B, Aging, Neuropsychology and Cognition, 22,* 220–243. http://dx.doi.org/10.1080/13825585.2014.911240

McClelland, J. L., & Rumelhart, D. E. (1981). An interactive activation model of context effects in letter perception: I. An account of basic findings. *Psychological Review, 88,* 375–407. http://dx.doi.org/10.1037/0033-295X.88.5.375

New, B., Brysbaert, M., Veronis, J., & Pallier, C. (2007). The use of film subtitles to estimate word frequencies. *Applied Psycholinguistics, 28,* 661–677. http://dx.doi.org/10.1017/S014271640707035X

New, B., Pallier, C., Brysbaert, M., & Ferrand, L. (2004). Lexique 2: A new French lexical database. *Behavior Research Methods, Instruments, & Computers, 36,* 516–524. http://dx.doi.org/10.3758/BF03195598

Perea, M. (2015). Neighborhood effect in visual word recognition and reading. In A. Pollatsek & R. Treiman (Eds.), *The Oxford handbook of reading* (pp. 76–87). New York, NY: Oxford University Press.

Perea, M., & Pollatsek, A. (1998). The effects of neighborhood frequency in reading and lexical decision. *Journal of Experimental Psychology: Human Perception and Performance, 24,* 767–779. http://dx.doi.org/10.1037/0096-1523.24.3.767

Perfetti, C. A. (2007). Reading ability: Lexical quality to comprehension. *Scientific Studies of Reading, 11,* 357–383. http://dx.doi.org/10.1080/10888430701530730

Schneider, W., Eschman, A., & Zuccolotto, A. (2002). E-Prime Reference Guide. (Version 2.0) [Computer software and manual]. Pittsburgh, PA: Psychology Software Tools Inc.

Sears, C. R., Campbell, C. R., & Lupker, S. J. (2006). Is there a neighborhood frequency effect in English? Evidence from reading and lexical decision. *Journal of Experimental Psychology: Human Perception and Performance, 32,* 1040–1062. http://dx.doi.org/10.1037/0096-1523.32.4.1040

Stanovich, K. E., & West, R. F. (1989). Exposure to print and orthographic processing. *Reading Research Quarterly, 24,* 402. http://dx.doi.org/10.2307/747605

van Heuven, W. J. B., Dijkstra, T., & Grainger, J. (1998). Orthographic neighborhood effects in bilingual word recognition. *Journal of Memory and Language, 39,* 458–483. http://dx.doi.org/10.1006/jmla.1998.2584

Yarkoni, T., Balota, D., & Yap, M. (2008). Moving beyond Coltheart's N: A new measure of orthographic similarity. *Psychonomic Bulletin & Review, 15,* 971–979. http://dx.doi.org/10.3758/PBR.15.5.971

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